

# RX111 Ultra Low Power, High Flash controller with USB OTG for THERMAL PRINTER

Thermal printer has increased in popularity because of its many advantages such as costs, printing speed, graphic capabilities, quality of print and mobility.

Renesas developed the USB thermal printer mini interface module to provide a high-quality, cost-effective and space-saving thermal printing solution. The module is developed with the RX111 32-bit high-performance, ultra low power consumption microcontrollers. The devices integrate a USB2.0 Host/Function controller, OTG communication peripheral, BC 1.2 Battery Charging and AoA support. Moreover, they come in wide ranging on chip memory size make them perfect for thermal printer solutions.



## Key Benefits

- Low Power Consumption 32 Bit MCU.
- Support BC 1.2 Charging from standard USB Port
  - Standard Mobile Charger / USB adapter for charging
  - Can charge Li-Ion battery from Power Bank.
- Firmware and Font Upgrade through USB.
- Support Multilingual fonts > 3 language fonts (Hindi, Kannada, English) support without any external memory due to large internal memory.
- Communication Interfaces
  - USB OTG
  - Serial
  - Bluetooth
- Supports AoA (Android open Source) can print through USB OTG
- High Speed Printing : 55-60mm/s, Text File content, Logo, Bar Code Support (Max length of 10 values Includes alphabets and numbers)

## Applications

- POS
- Ticketing terminal
- Taxi meter
- KIOSK
- Labeling
- Banking
- Medical devices
- Instrumentation and etc

## Specification

- Paper and platen Detect options with LED indication
- LED to indicate print process (on Berg)
- Battery: Li Ion (7.4V, 1100mAh)
- Supported Printers: 2 inch

### Memory

Zero-wait Flash  
512KB

SRAM  
64KB

Data Flash  
8KB

### RX 32-bit CPU (3.8 CoreMark/MHz)

#### DSP Instructions

Register Indirect Multiply-and-Accumulate  
(Result: 80-bit)

Register Indirect Multiply-and-Accumulate  
(Result: 48-bit)

Barrel Shifter: 32-bit

### System

Event Link  
Controller

Multi-Pin Function  
Controller

Data Transfer  
Controller

Interrupt Control  
16 levels, 9 pins

Clock Generation  
Circuit  
High-speed  
On-chips Oscillator

Power-On Reset  
Voltage Detection  
Circuit

Clock Frequency  
Accuracy  
Measurement Circuit  
Data Operation Circuit  
CRC Calculator

### Timers

Multi-Function  
Timer Pulse Unit  
16-bit x 6 ch

Compare Match  
Timer  
16-bit x 2 ch

14-bit Independent  
Watchdog Timer

Real-Time Clock  
Calendar Function

### Communication Functions

USB x 1 ch

I<sup>2</sup>C Bus Interface

Serial  
Communications  
Interface x 3 ch

Serial Peripheral  
Interface

### Analog

12-bit A/D x 14 ch

8-bit A/D x 2 ch

Temperature Sensor



बायल भाषा  
मलयालम  
तमिल  
कन्नडा  
हिन्दी  
उर्दू  
बंगाली  
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